Process Analysis Report

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Executive Summary

The simulations indicate that the temperature of Air, Fuel, and the cold fluid in HEX-100 are the most significant factors influencing the Heater Outlet Temperature. Optimizing these temperatures is critical to achieving the desired KPI performance. Focus should be given to the Air temperature given it's high weightage compared to other variables.

Top Variables Analysis

To achieve the top 5% of the KPI, which is to maximize the Heater Outlet Temperature, the focus should be on precisely controlling the setpoints of Air temperature, Fuel temperature, and the cold fluid temperature within the HEX-100 heat exchanger. The impact weightages of 47.903, 27.853, and 24.244 respectively, indicate the relative importance of these variables. While the HEX-100 condition (global heat transfer coefficient) appears to play a role in performance, by far the dominant impact can be found in temperature optimization.

Impact Breakdown

The setpoint impact is broken down as follows: Air temperature accounts for 47.903%, Fuel temperature accounts for 27.853%, and cold fluid temperature in HEX-100 accounts for 24.244% of the total impact on the KPI. The condition impacts from global_heat_transfer_coefficient, which is dependent on fluid flow and fouling, also plays a role; however, the available data does not have enough variances and weightage reporting to create recommendations. As conditions degrade (scaling, corrosion), the importance of temperatures will increase and likely require higher temperatures to maintain KPI performance.

Recommendations

1. Implement tighter control loops on Air, Fuel, and HEX-100 cold fluid temperature setpoints to minimize deviations from optimal values.

2. Investigate the cost benefit of upgrading the air and fuel supply streams to improve temperature control and stability and lower the likelihood of out-of-spec temperatures.

3. Develop real-time optimization strategies that dynamically adjust temperature setpoints based on the global heat transfer coefficient of the HEX-100 to compensate for fouling or changes in heat transfer efficiency.

4. Focus data collection efforts on gaining more details on how the global_heat_transfer_coefficient interacts with the temperature setpoints so that a future report can

quantify the interactions.

Data Tables

Top Impact Variables and Their Setpoints

Shows the equipment, variable type, variable name, value, and unit for each of the top variables that impact the KPI.

Equipment	Type	Name	Value	Unit
HEX-100	Setpoint	cold_fluid_temperature	329.974	K
Fuel	Setpoint	temperature	364.524	K
Air	Setpoint	temperature	305.88	K

Setpoint Impact Weightages

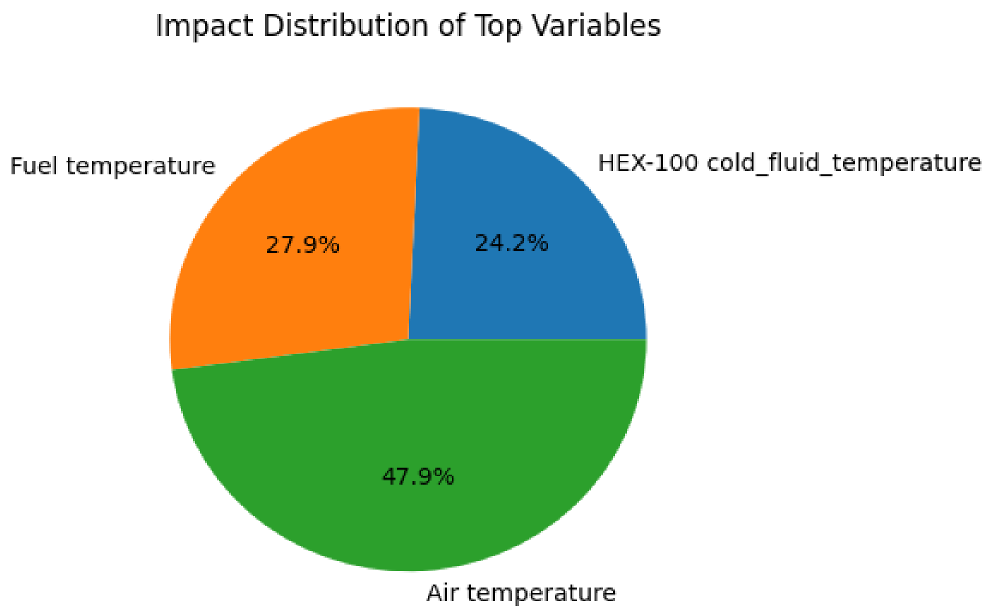
Displays the weightage assigned to each setpoint based on its impact on the KPI.

Equipment	Setpoint	Weightage	Unit
HEX-100	cold_fluid_temperature	24.244	K
Fuel	temperature	27.853	K
Air	temperature	47.903	K

Data Visualizations

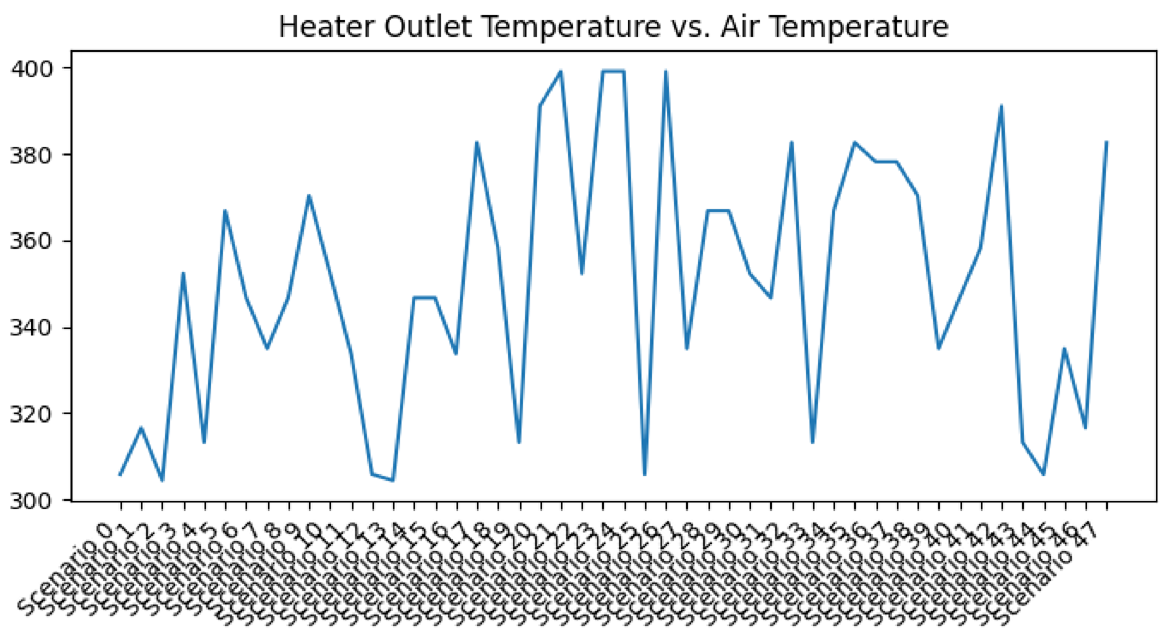
Impact Distribution of Top Variables

Illustrates the percentage contribution of each variable to the overall KPI impact.



Heater Outlet Temperature vs. Air Temperature

Shows the relationship between Air Temperature and Heater Outlet Temperature across all Scenarios



KPI Summary Across Scenarios

The analysis includes 48 scenarios with KPI values. The average KPI value across all scenarios is 381.17. The maximum KPI value is 500.89 and the minimum is 328.60.

Scenario	KPI Value	% of Maximum
Scenario 0	500.89	100.0%
Scenario 1	461.34	92.1%
Scenario 2	441.18	88.1%
Scenario 3	433.88	86.6%
Scenario 4	416.69	83.2%
Scenario 5	409.46	81.7%
Scenario 6	404.91	80.8%
Scenario 7	400.16	79.9%
Scenario 8	396.72	79.2%
Scenario 9	393.66	78.6%
Scenario 10	393.44	78.5%
Scenario 11	393.34	78.5%
Scenario 12	391.41	78.1%
Scenario 13	390.17	77.9%
Scenario 14	386.75	77.2%
Scenario 15	386.57	77.2%
Scenario 16	385.81	77.0%
Scenario 17	382.74	76.4%
Scenario 18	381.37	76.1%
Scenario 19	381.07	76.1%
Scenario 20	380.18	75.9%
Scenario 21	379.84	75.8%
Scenario 22	378.16	75.5%
Scenario 23	376.88	75.2%
Scenario 24	372.94	74.5%
Scenario 25	369.93	73.9%
Scenario 26	367.87	73.4%
Scenario 27	367.73	73.4%
Scenario 28	367.66	73.4%
Scenario 29	367.43	73.4%
Scenario 30	366.07	73.1%
Scenario 31	365.87	73.0%
Scenario 32	365.37	72.9%

Scenario 33	363.75	72.6%
Scenario 34	362.21	72.3%
Scenario 35	361.46	72.2%
Scenario 36	360.51	72.0%
Scenario 37	359.70	71.8%
Scenario 38	359.42	71.8%
Scenario 39	359.35	71.7%
Scenario 40	357.68	71.4%
Scenario 41	356.86	71.2%
Scenario 42	356.77	71.2%
Scenario 43	356.14	71.1%
Scenario 44	354.29	70.7%
Scenario 45	351.30	70.1%
Scenario 46	350.77	70.0%
Scenario 47	328.60	65.6%